Science of the Total Environment 778 (2021) 146129

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

New assessment for residential greenness and the association with cortical thickness in cognitively healthy adults

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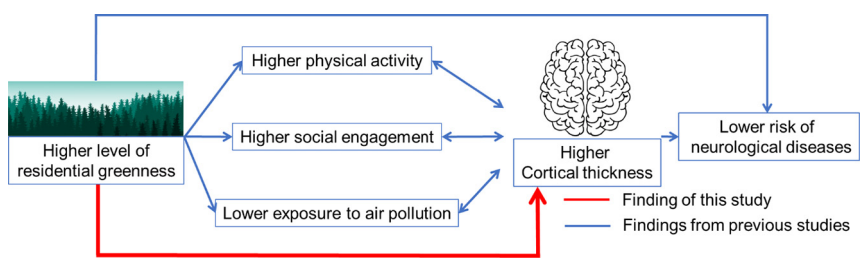
HIGHLIGHTS

- Higher residential greenness was associated with higher cortical thickness in an urban population
- Effect estimates increases with increased buffer sizes to define residential greenness in an urban population
- No association was found between cortical thickness and residential greenness in non-urban populations

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 23 September 2020
Received in revised form 26 January 2021
Accepted 22 February 2021
Available online 1 March 2021

Editor: Lotfi Aleya

Keywords:
Brain imaging
Cortical thickness
Enhanced vegetation index
Neurodegenerative disorder
Residential greenness

ABSTRACT

Background: Recent evidence suggests that neurological health could be improved with the intervention of local green space. A few studies adopted cortical thickness, as an effective biomarker for neurodegenerative disorder, to investigate the association with residential greenness. However, they relied on limited data sources, definitions or applications to assess residential greenness. Our cross-sectional study assessed individual residential greenness using an alternative measure, which provides a more realistic definition of local impact and application based on the type of area, and investigated the association with cortical thickness.
Methods: The study population included 2542 subjects who participated in the medical check-up program at the Health Promotion Center of the Samsung Medical Center in Seoul, Korea, from 2008 to 2014. The cortical thickness was calculated by each of the four and global lobes from brain MRI. For greenness, we used the enhanced vegetation index (EVI) that detects canopy structural variation by adjusting background noise based on satellite imagery data. To assess individual exposure to residential greenness, we computed the maximum annual EVI before the date of a medical check-up and averaged it within 750 m from subjects' homes to represent an average walking distance. Finally, we assessed the association with cortical thickness by urban and non-urban populations using multiple linear regression adjusting for individual characteristics.
Results: The average global cortical thickness and EVI were 3.05 mm (standard deviation = 0.1 mm) and 0.31 (0.1), respectively. An interquartile range increase in EVI was associated with 11 μm (95% confidence interval = 3–20) and 9 μm (1–16) increases in cortical thickness of the parietal and occipital regions among the urban population. We did not find associations in non-urban subjects.

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https://doi.org/10.1016/j.scitotenv.2021.146129
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Conclusions: Our findings confirm the association between residential greenness and neurological health using alternative exposure assessments, indicating that high exposure to residential greenness can prevent neurological disorders.

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1. Introduction

Cortical thickness is indicated as a clinically important biomarker that is associated with clinical symptoms or diseases in the earlier stage of neurodegenerative disorders. Cortical thickness is a distance measurement between white matter surfaces and pial surfaces, and it implies the number of neurons within a cortical column (Burgaleta et al., 2014; Cordova-Palomera et al., 2014; Rakic, 1988). There has been growing evidence that cortical thickness is negatively associated with various neurological symptoms and diseases, such as depression, cognitive function and dementia (Pacheco et al., 2015; Peterson et al., 2009; Zarei et al., 2013). Studies in the disciplines of neurology and neural science have suggested various associated risk factors for cortical thinning, mostly focusing on individual characteristics. Cortical thinning was associated with higher body mass index (Shaw et al., 2018), hypertension (Gonzalez et al., 2015), chronic stress (Lucassen et al., 2014; Radley et al., 2015) and shorter duration of physical activity (Lee et al., 2016; Rogge et al., 2018). Social and environmental factors related to cortical thickness were also suggested. Voluntary prosocial behaviors were associated with the regional development of cortical thickness in adolescents and young adults (Ferschmann et al., 2019), while social support was positively associated with prefrontal cortical thickness in older adults (Sherman et al., 2016). Exposure to air pollution was also examined in both paediatric (Beckwith et al., 2020) and elderly populations (Casanova et al., 2016), showing a negative association with cortical thickness. However, these results were inconsistent in other studies (Chen et al., 2015; de Prado Bert et al., 2018).

Residential greenness is defined as the level of visual greenness around residences and is considered as a proxy for accessibility to public parks or residential gardens (Ren et al., 2017). Residential greenness may be another environmental factor related to cortical thickness. Previous findings of associations between greenness and the risk factors of cortical thickness suggest the possibility of a direct association between greenness and cortical thickness. A high level of residential greenness was associated with stress mitigation (Astell-Burt et al., 2010; de Vries et al., 2013), increased motivation for social engagement (de Vries et al., 2013; Sugiyama et al., 2008), increased physical activity (Astell-Burt et al., 2010; Grigsby-Toussaint et al., 2011; Lachowycz et al., 2012; Sadeh et al., 2019) and reduced adverse impact of air pollution (Wang et al., 2020). The changes in these characteristics subsequently affected cortical thickness (Beckwith et al., 2020; Casanova et al., 2016; Chen et al., 2015; de Prado Bert et al., 2018; Ferschmann et al., 2019; Lee et al., 2016; Rogge et al., 2018; Sherman et al., 2016). In addition, residential greenness was associated with various neurological symptoms and diseases, such as depression (Min et al., 2017; Song et al., 2019), frailty among elderly (Zhu et al., 2019b) and cognitive functions (de Keijzer et al., 2018; Zhu et al., 2019a), which were also associated with cortical thickness (Jiang et al., 2016; Pacheco et al., 2015; Peterson et al., 2009; Zarei et al., 2013).

Recently, an association between greenness and brain structure has been reported in a few studies. A Spanish study used a satellite-based normalized difference vegetation index (NDVI) to assess residential greenness exposure in 253 school children in Barcelona and found a positive association in grey and white matter (GM and WM) volume (Dadvand et al., 2018). Another study in Barcelona used 958 cognitively unimpaired subjects aged 45–74 years and showed a positive association between NDVI and Alzheimer's disease-related regional cortical thickness measurements including those of the entorhinal, inferior and middle temporal, fusiform, posterior cingulate, precuneus and supramarginal gyri (Crous-Bou et al., 2020). A study of 112 middle-aged Bulgarians also used residential NDVI and found a positive association with average cortical thickness across both hemispheres (Dzhambov et al., 2019). All these studies, however, relied on similar exposure assessments.

Additional studies that adopt an alternative exposure measure and application can bolster the association between residential greenness and cortical thickness. First, an alternative measure for residential greenness needs to be applied to confirm the association. Although NDVI has been widely used for assessing greenness in health studies (James et al., 2015), the enhanced vegetation index (EVI) can be a complementary and possibly more sensitive measure to detect canopy structural variation as it adjusts background noise due to aerosol resistance (Huete et al., 2002). Despite its strengths, EVI has rarely been applied to epidemiological studies except for in a few examples. A recent U.K. study investigated the effect of residential surrounding greenness on physical activity using NDVI as well as EVI (de Keijzer et al., 2019). Second, the distance used to assess residential greenness could be extended. Previous studies used relatively small circular buffers with radii of 100 (Dadvand et al., 2018; Dzhambov et al., 2019) to 300 m (Crous-Bou et al., 2020) from participants' homes to define the area within access to residential greenness. However, walkable distances to greenness such as public parks could be longer and be represented by larger buffer sizes. Third, greenness may have different implications based on the type of area. Greenness is represented by widely variant forms of green space including parks, public gardens, sports fields, forests, open land and cropland (Han and Kim, 2019). Predominant forms of green space could differ for urban and rural areas, leading to different impacts on physical activity, psychological reactions and health outcomes. In a previous study, greenness was associated with frailty among older populations in urban areas but not in rural areas (Zhu et al., 2019b). In addition, the intelligence quotient in 6-year-old children was associated with artificial greenness, but not with wild greenness (Lee et al., 2020).

Taking advantage of a larger study population of more than 2500 cognitively healthy adults who participated in a preventive medical screening program in South Korea, we applied a novel approach to assess residential greenness based on EVI and walkable distance as assessed in urban or rural areas, and investigated the association with cortical thickness.

2. Materials and methods

2.1. Data acquisition and preprocessing

2.1.1. Study population

We constructed a cross-section study using a total of 3333 individuals who participated in a preventive medical screening program at the Health Promotion Center of the Samsung Medical Center in Seoul, Korea, from September 2008 to December 2014. SMC is one of the largest hospitals in South Korea, and it provides a high-quality screening program. Participants from all over the country desired to join, although this screening is not covered by national health insurance. The 3333 individuals included cognitively healthy adults based on the Christensen criteria (Christensen et al., 1991) and other assessments. We excluded those who had histories of psychiatric and/or neurologic illnesses, and abnormal symptoms and/or signs in neurologic examinations suggesting underlying neurologic illnesses. Also, we excluded those who showed a Korean-Mini Mental Status Examination (K-MMSE) score

below the 16th percentile (Kang et al., 2019) and structural lesions such as territorial infarction, hemorrhage, tumor and injury in their brain magnetic resonance imaging (MRI). All the subjects underwent brain MRI including three-dimensional (3D) volume images. Fig. 1 shows our application of additional exclusion criteria to finalize the study population. First, we excluded subjects with the following conditions: (1) 291 subjects with unreliable cortical thickness data due to head motion, blurring of the MRI, inadequate registration to a standardized stereotaxic space, misclassification of tissue type or inexact surface extraction and (2) 423 subjects who had incomplete information on demographics ($N = 79$), comorbidity including hypertension, diabetes, hyperlipidemia and obesity ($N = 26$) and unhealthy behavior such as smoking and drinking ($N = 318$). Additional exclusion was applied to those who had incomplete EVI data ($N = 74$) and were aged under 45 years (N = 3). Overall, 2542 subjects were included in this study.

This study was approved by the Institutional Review Board of the Samsung Medical Centre (SMC 2020-06-169). The requirement for informed consent was waived because we only used unidentifiable data collected for clinical purposes during health screening exams.

2.1.2. Health screening and questionnaire

The health screening exams were performed by trained personnel according to standard protocols, as previously described in detail (Lee et al., 2016; Park et al., 2010). A standardized questionnaire was used to collect information related to demographic characteristics, total duration of formal education, medical history (including medication intake) and health-related social history such as alcohol use and smoking. Hypertension was defined as systolic blood pressure ≥ 140 mm Hg, diastolic blood pressure > 90 mm Hg, a self-reported history of hypertension or current use of antihypertensive medication. Diabetes was defined as a fasting serum glucose level > 126 mg/dL, self-reported diabetes history or use of diabetes medications. Dyslipidemia was defined as fasting total cholesterol > 200 mg/dL, low density lipoprotein > 100 mg/dL, high density lipoprotein < 50 mg/dL, triglyceride > 200 mg/dL, self-reported history of dyslipidemia or current use of lipid-lowering medication. Obesity was defined as a body mass index greater than 25 kg/m^2.

2.1.3. Brain MRI scan

All subjects underwent 3D volumetric brain MRI scans. The 3D T1 Turbo Field Echo (TFE) MRI data were acquired through a sagittal slice thickness of 1.0 mm, contiguous slices with 50% overlap and no gap, a repetition time of 9.9 ms, an echo time of 4.6 ms, a flip angle of 8° and a matrix size of 240 × 240 pixels reconstructed to 480 × 480 over a field of view of 240 mm using an Achieva 3.0-Tesla MRI scanner (Philips, Best, the Netherlands). Radiologists initially inspected all MR images for evidence of brain tumors, lobar infarctions (except lacunar infarctions) and hemorrhages (observed as low-intensity areas in T2-weighted images). T1-weighted MR images were processed automatically using the standard Montreal Neurological Institute image processing software, CIVET, to measure cortical thickness. The software has been well-validated and described extensively elsewhere including aging/atrophied brain studies (Kabani et al., 2001; Lee et al., 2006; Lerch and Evans, 2005; Lerch et al., 2005; Singh et al., 2006). In summary, native MRI images were first registered into a standardized stereotaxic space using an affine transformation (Collins et al., 1994). Non-uniform artefacts were corrected using the N3 algorithm, and the registered and corrected volumes were classified as white matter, GM, cerebrospinal fluid (CSF) and background using an artificial neural net classifier (Sled et al., 1998). The surfaces of the inner and outer cortices were extracted automatically by deforming a spherical mesh onto the grey/white boundary of each hemisphere using the constrained Laplacian-based automated segmentation with a proximities algorithm, which has also been well-validated and extensively described elsewhere (Kim et al., 2005; MacDonald et al., 2000).

2.1.4. Cortical thickness

Cortical thickness was calculated as the Euclidean distance between the linked vertices of the inner and outer surfaces after applying an inverse transformation matrix to cortical surfaces and reconstructing them in the native space (Im et al., 2006; Kim et al., 2005). To compare the thicknesses of corresponding regions among subjects, the thicknesses were spatially registered on an unbiased iterative group template by matching the sulcal folding pattern using surface-based registration involving sphere-to-sphere warping (Lyttelton et al., 2007). For global and lobar regional analyses, we used the lobe-parcellated group template that had been previously divided into frontal, temporal, parietal and occipital lobes using SUMA software (http://afni.nimh.nih.gov) (Im et al., 2006). The cortical thickness of each lobe was calculated as an average of the whole vertex in each lobar region. Global cortical thickness was the average of the cortical thickness of each lobe.

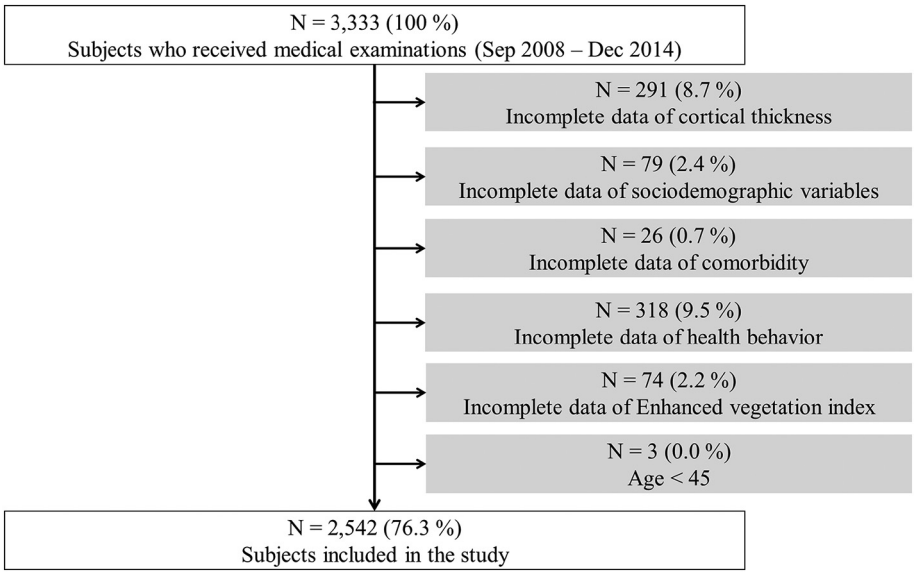

Fig. 1. The numbers of study subjects included as the final study population and excluded in the selection procedure based on exclusion criteria.

2.1.5. Exposure to residential greenness

We used EVI to assess the residential vegetation level of study subjects. EVI is a ratio that includes near-infrared (NIR), which scatters in leaf mesophyll cells, and red reflectance (Red) absorbed by photosynthetic chlorophyll. Therefore, the index is considered as a proxy for the greenness of the vegetation canopy. Compared to NDVI, which is computed using the two wavelengths of NIR and Red (Eq. (1)), EVI additionally includes adjustment factors for canopy background (L) and aerosol resistance (C_1, C_2) and a gain factor (G) (Eq. (2)). L accounts for the difference in Red and NIR extinction in the dense vegetation area where Red extinction results in decreasing sensitivity of the vegetation index (Huete et al., 1997). C_1 and C_2 are the coefficients to correct the red band for aerosol influence, which is accomplished by excluding the atmosphere-sensitive blue band. Therefore, EVI is more responsive to canopy structural variations and provides improved vegetation monitoring through a reduction in atmospheric influence (Huete et al., 2002).

$$NDVI = \frac{NIR - Red}{NIR + Red} \tag{1}$$

$$EVI = G \times \frac{NIR - Red}{NIR + C_1 \times Red - C_2 \times Blue + L} \tag{2}$$

The National Aeronautics and Space Administration (NASA) Moderate Resolution Imaging Spectroradiometer (MODIS) database (https://terra.nasa.gov/about/terra-instruments/modis) provides global EVI data with a temporal resolution of 16 days and spatial resolution of 250 m. Using the 'MODIStsp' package (Busetto and Ranghetti, 2016) in R v.3.5.1. (R Core Team, 2013) (The R Foundation for Statistical Computing, Vienna, Austria; https://www.r-project.org/), we downloaded raster type EVI data for all of South Korea between 2007 to 2014 including 23 sets of 16-day data files each year. For individual greenness exposure of each subject, we calculated the annual maximum EVI for one year before the date of MRI scanning on each grid and averaged it within a 750-m circular buffer from the home of each subject. The 750-m buffer was applied based on a previous study that reported an average walking distance per day of 744 m in South Korea (Oh, 2014). We also computed average EVIs within 250-, 500- and 1000-m buffers for our sensitivity analysis, as smaller and larger effective areas, to compare with the 750-m buffer in our primary analysis. The 'Raster' package (Hijmans et al., 2015) in R v.3.5.1. was used to extract EVI values for different buffer areas. In addition, to examine the sensitivity to different vegetation indices, we used NDVI and the soil-adjusted vegetation index (SAVI) (Huete, 1988). SAVI corrects NDVI for the influence of light reflected from soil (L), especially in areas where greenness is low (Eq. (3)).

$$SAVI = \frac{NIR - Red}{NIR + Red + L} * (1 + L) \tag{3}$$

2.2. Statistical analysis

We conducted descriptive analyses of subjects' general characteristics such as age, sex, duration of formal education, smoking and drinking behaviors, and comorbidities. The mean and standard deviation of cortical thickness of subjects and their EVI levels were also described. To investigate whether the association between residential greenness and cortical thickness was different based on the type of area, we performed separate analyses by urban and non-urban districts. To this end, we acquired a list of all 250 districts in South Korea classified into 106 urban, 59 suburban and 85 rural districts from the Ministry of the Interior and Safety (2017). The classification was re-categorized to urban and non-urban areas because relatively small numbers of subjects lived in suburban and rural districts. We performed the Pearson's chi-square test and the Student's *t*-test to examine the differences in general characteristics between urban and non-urban areas.

To examine the association between residential greenness and cortical thickness, we applied multivariable linear regression adjusting for individual characteristics. Each of these characteristics was associated with cortical thickness in previous studies (Alosco et al., 2014; Brundel et al., 2010; Cho et al., 2016; Fjell et al., 2009; Fortier et al., 2011; Karama et al., 2015; H. Kim et al., 2015; J.P. Kim et al., 2015; Kim et al., 2019; Lee et al., 2018; Meyer et al., 1999; Veit et al., 2014). We used categorized age in our health analysis, because the subjects equal to or over 65 years old tend to have higher risk to have cognition impairment. The analysis was repeated for five sets of average cortical thickness from the left and right hemispheres: a global measure and four regional measures for frontal, parietal, temporal and occipital lobes. We presented a difference in cortical thickness for an interquartile range (IQR) increase in terms of annual maximum EVI with its 95% confidence interval (CI). In addition, we used quartiles of annual maximum EVI to represent a potential non-linear association with cortical thickness. Using the lowest quartile (1st quartile) as a reference category, we presented the differences in cortical thickness from each of the other three quartiles. All analysis was stratified by urban and non-urban areas and we additionally conducted a statistical analysis including an interaction term (EVI × urban/non-urban categorization) in the total population. In our sensitivity analysis, we used smaller and larger buffer sizes of 250, 500 and 1000 m and compared the results with our primary findings using the 750-m buffer. We also replaced EVI by NDVI and SAVI and compared the results to our original findings using EVI.

All statistical analyses conducted in this study were performed using R v 3.5.1.

3. Results

The general characteristics of study subjects are described in Table 1. About 70% of subjects were aged more than 65 years (The mean age of our subjects was 68, ranged from 45 years old to 96 years old), while approximately half of the population were men and obtained more than or equal to 15 years of formal education. Never-smokers and never-drinkers made up 61 and 50%, respectively. Prevalence of hypertension, diabetes, hyperlipidemia and obesity were 45.8, 15.7, 33.1 and 33.2%, respectively. These general characteristics were not significantly different between urban and non-urban districts, except for age and educational obtainment. Mean EVI in non-urban areas was 0.40 and significantly higher than 0.28 in urban areas. Mean cortical thickness was 3.05, 3.10, 3.21, 2.91 and 2.69 mm, for global, frontal, temporal, parietal and occipital regions, respectively. None of the five sets of cortical thickness were significantly different between the two types of areas.

Using the annual maximum EVI, estimates for the associated difference in cortical thickness were positive in all lobes except for the temporal region (Fig. 2). However, none of these effect estimates were statistically significant. When our analysis focused on the populations in urban districts, an IQR increase in EVI was significantly or marginally associated with 7 μm (95% CI = 0–13 μm, *p* value = 0.038), 11 μm (3–20 μm, 0.009), 9 μm (1–16 μm, 0.021) and 6 μm (−1–13 μm, 0.074) increases in global, parietal, occipital and frontal cortical thickness, respectively. In contrast, we did not find any association in non-urban districts. Effect estimates were close to zero or negative with a significantly negative association for an occipital cortical thickness decrease (15 μm [95% CI = 29–1 μm, p value = 0.039]) for an IQR increase in EVI. The analysis with total population including an interaction term (EVI × urban/non-urban categorization) showed that the associations of EVI were significantly different with parietal and occipital thickness (*p* values = 0.023 and 0.002, respectively) between urban and non-urban regions. The different patterns of the association by urban and non-urban districts were consistent when we used quartiles of EVI. In the urban population, the 2nd and 4th quartiles of EVI were associated with 16 μm (95% CI = 3–30 μm, p value = 0.021) and 19 μm (6–33 μm, 0.006) increases, respectively, in global cortical thickness compared to the 1st (lowest) quartile (Fig. 3). These positive

Table 1
Summaries of cortical thickness, Enhanced Vegetation Index and general characteristics of study subjects.

Variables	Levels	Total		Urban		Non-urban		Univariable test*
		N	%	N	%	N	%	χ^2 test
Total		2542	100	1891	100	651	100	
Age	<65	777	30.6	557	29.5	220	33.8	0.043
	≥65	1765	69.4	1334	70.5	431	66.2	
Sex	Men	1264	49.7	933	49.3	331	50.8	0.537
	Women	1278	50.3	958	50.7	320	49.2	
Education	≥15 years	1129	44.4	922	48.8	207	31.8	<0.001
	<15 years	1413	55.6	969	51.2	444	68.2	
Smoke	Never	1560	61.4	1157	61.2	403	61.9	0.522
	Past	765	30.1	578	30.6	187	28.7	
	Current	217	8.5	156	8.2	61	9.4	
Drink	Never	1257	49.4	926	49.0	331	50.8	0.513
	Past	1283	50.5	963	50.9	320	49.2	
	Current	2	0.1	2	0.1	0	0.0	
HTN[a]	Yes	1164	45.8	871	46.1	293	45.0	0.675
	No	1378	54.2	1020	53.9	358	55.0	
DM[b]	Yes	399	15.7	295	15.6	104	16.0	0.869
	No	2143	84.3	1596	84.4	547	84.0	
HL[c]	Yes	841	33.1	638	33.7	203	31.2	0.251
	No	1701	66.9	1253	66.3	448	68.8	
Obesity	Yes	844	33.2	614	32.5	230	35.3	0.198
	No	1698	66.8	1277	67.5	421	64.7	
		Mean	SD	Mean	SD	Mean	SD	*t*-Test
EVI[d]		0.31	0.1	0.28	0.1	0.40	0.1	<0.001
Cth[e] (mm)	Global	3.05	0.1	3.05	0.1	3.05	0.1	0.447
	Frontal	3.10	0.1	3.10	0.1	3.10	0.1	0.267
	Temporal	3.21	0.2	3.21	0.2	3.21	0.2	0.710
	Parietal	2.91	0.1	2.91	0.1	2.92	0.1	0.581
	Occipital	2.69	0.1	2.69	0.1	2.69	0.1	0.707

Note: Study subjects were 2542 cognitively healthy adults who participated in a preventive medical screening program at the Health Promotion Center of the Samsung Medical Center in Seoul, Korea, from September 2008 to December 2014.
[a] Hypertension.
[b] Diabetes mellitus.
[c] Hyperlipidemia.
[d] Enhanced vegetation index.
[e] Mean cortical thickness.
* Univariable test was conducted to examine the difference in each variable between urban and non-urban areas.

associations with global cortical thickness were consistent with slightly increased effect estimates in parietal (difference = 23 μm [95% CI = 5–41 μm, p value = 0.012] and 28 μm [11–46 μm, 0.002] for the 2nd and 4th quartiles, respectively), occipital (18 μm [2–34 μm, 0.023] and 23 μm [7–39 μm, 0.004]) and frontal regions (15 μm [0–29 μm, 0.047] and 18 μm [4–33 μm, 0.015]).

The results were generally consistent in our sensitivity analysis using 250-, 500- and 1000-m buffers instead of a 750-m buffer, but

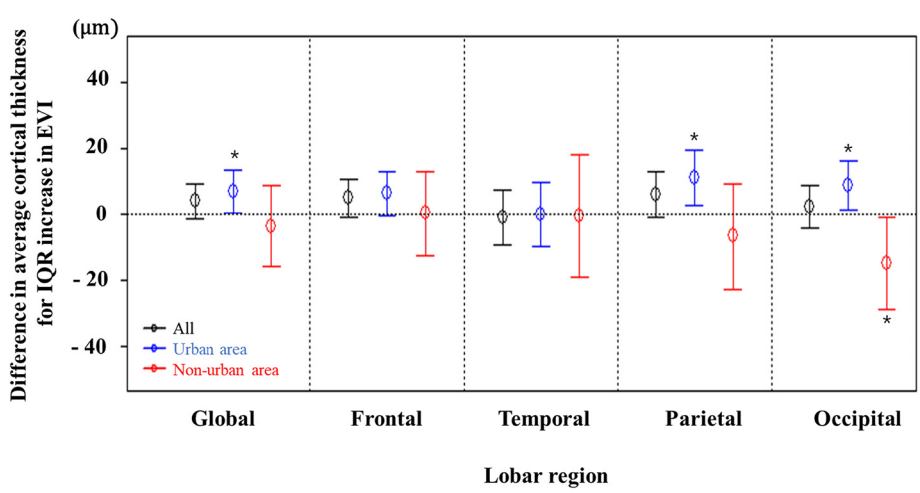

Fig. 2. Estimated differences in average cortical thickness in each of the global and four lobar regions for an interquartile range increase in annual maximum enhanced vegetation index (EVI) for the previous one year before MRI scanning averaged within 750 m from subjects' homes after adjusting for individual characteristics (age, sex, duration of education, smoking, drinking, hypertension, diabetes, hyperlipidemia and obesity) by all ($N = 2542$), urban ($N = 1891$), and rural ($N = 651$) districts in 2542 cognitively healthy adults who visited the Health Promotion Center of the Samsung Medical Center in Seoul, Korea, from September 2008 to December 2014 (* indicates statistical significance).

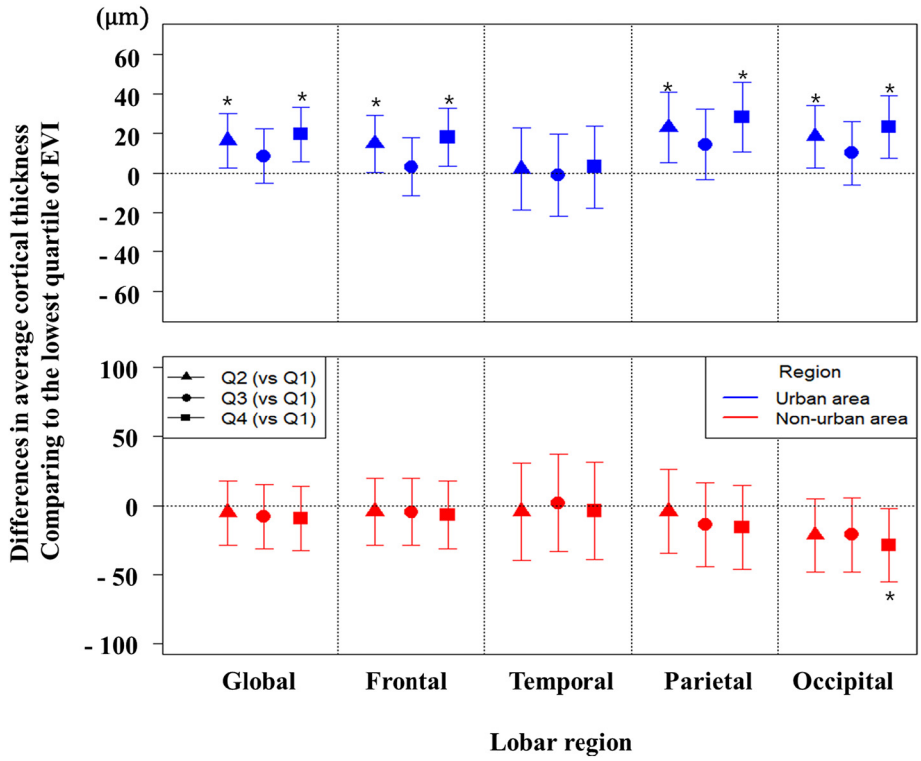

Fig. 3. Estimated differences in cortical thickness in each of the global and four lobar regions between the lowest quartile (Q1) and each of the second to fourth quartiles (Q2–Q4) of annual maximum enhanced vegetation index (EVI) for the previous one year before MRI scanning averaged within 750 m from subjects' homes after adjusting for individual characteristics (age, sex, duration of education, smoking, drinking, hypertension, diabetes, hyperlipidemia and obesity) by urban (N = 1891) and rural (N = 651) districts in cognitively healthy adults who visited the Health Promotion Center of the Samsung Medical Center in Seoul, Korea, from September 2008 to December 2014 (* indicating statistical significance).

with slightly increased effect estimates with larger buffer sizes (Figs. S1–S3). The effect estimates tended to be toward null when the 500-m and 250-m buffers were applied. Significantly positive effect estimates in global and occipital regions among the urban population found in the primary analysis using 750-m were not replicated, although the positive association in the parietal region remained. In contrast, the positive association among the urban population became more evident with increased effect estimates in the analysis using the 1000-m buffer. These patterns were also consistent with those for quartiles of EVI. However, while the 2nd and/or 4th quartiles showed significantly higher cortical thickness compared to the 1st quartile among the urban population based on the 750-m buffer, the 3rd and 4th quartiles gave significant increases in the 1000-m buffer. When NDVI and SAVI were used instead of EVI, the results were generally similar to primary findings (Figs. S4–S6). The direction and significance of the association of cortical thickness for SAVI were the same to those for EVI. NDVI tended to give higher effect estimates, particularly for temporal cortical thickness in the urban population.

4. Discussion

This study examined the association between residential greenness and cortical thickness using the average EVI within a walkable distance of 750 m from the homes of 2542 cognitively healthy adults living in urban and non-urban areas. Our findings showed a positive association between residential greenness and cortical thickness in parietal and occipital regions among the urban population. When we used EVI quartiles, cortical thickness in the 2nd and 4th quartiles was higher than in the lowest quartile with similar effect estimates among the 2nd to 4th quartile. However, this positive association was not found among the non-urban population. The positive association between exposure to residential greenness and cortical thickness among the urban population was more evident when we used larger buffer sizes to define exposure to residential greenness. These results were generally consistent when we used NDVI and SAVI as alternative vegetation indices.

Our findings based on cortical thicknesses measured in about 2500 healthy people (as a subclinical outcome of neurological diseases) indicate the positive effects of residential greenness on the early stages of neurological disorder. Previous studies on the potential beneficial effects of residential greenness mostly relied on the outcomes of later-stage disorders such as neurological diseases or self-reported symptoms (Lee and Maheswaran, 2011). Although a few studies used cortical thickness to investigate the association, the size of the study population was limited to a few dozens to hundreds (25–253 people) (Crous-Bou et al., 2020; Dadvand et al., 2018; Dzhambov et al., 2019). Our confirmation of an association using a much larger population and accompanied with alternative exposure assessments for residential greenness suggests the importance of residential greenness as a modifiable factor for neurological and mental health and has public health implications. The accumulation of evidence for the beneficial effect of greenness may facilitate creating green space in residential areas as a health promotion strategy.

The positive association in frontal, parietal and occipital regions among the urban population in our study could be explained by region-specific brain functions related to physical activity and social engagement, which can be promoted by accessibility to green space (Astell-Burt et al., 2010; de Vries et al., 2013; Sugiyama et al., 2008). A previous study reported that increased exercise enhanced cerebral blood flow of the parietal lobe (Robertson et al., 2017), indicating the subsequent prevention of cortical thinning. In addition, social engagement, such as the number of engagements for daily activities, was associated with increased grey matter volume in the occipital region (James et al., 2012). Previous findings of frontal lobe function related to physical activities such as walking (Lee et al., 2016; ten Brinke et al., 2018) also suggest the possibility that cortical thickening associated with easy access to green space is mediated by enhanced physical activity. Future

studies should examine the mechanisms of the association focusing on the role of potential mediators such as physical activity, social engagement, air pollution and noise in a longitudinal study design.

As opposed to the urban population, we did not find any evidence of an association in the rural population. Effect estimates were even significantly negative in the occipital region. There are two possible explanations. First, urban residents could be more reactive to the exposure to greenness even in low levels because they have been exposed to much less residential greenness compared to non-urban residents. In our study, the mean EVI in urban areas is about two thirds of that in non-urban areas (0.28 vs. 0.40). Second, residential greenness can have different implications in urban and rural areas. The higher level of residential greenness in urban areas could indicate the presence of public parks or urban trails which can improve physical activity, social engagement and/or visual stimulation. In contrast, it could be isolated areas, which reduce walkability and social capital for residents in non-urban areas. The effect of greenness may not uniformly correspond to the quantified level of greenness and could rather vary by the type of land use, which may be related to physical activities or social engagement. However, greenness in isolated non-urban areas could still have a protective impact via sight-seeing. For example, a previous study showed a positive health effect from virtual stimulation of greenness without any physical activity (White et al., 2018). Future studies should clarify the benefits and disadvantages related to greenness depending on the different types of areas. The different associations between urban and non-urban populations may also be due to different effects on population subgroups and different population compositions between urban and rural areas. However, our additional analysis that was stratified by age (greater than 65 or not) and sex showed similar effect estimates between subgroups (Fig. S7).

The positive association among the urban population was more prominent with larger buffer sizes to assess the exposure to residential greenness. This finding may represent the walking characteristics of the urban population in South Korea. Although the average travel distance by walking was 744 m in Seoul based on a travel diary survey, the average walking distance in leisure time was 950 m (Oh, 2014). In addition, the average travel time from residential locations to urban parks was reported to be 13.3 min (Kim et al., 2014), which is about 867 m assuming that the walking speed is 4 km per hour. Although a 500-m buffer was commonly applied in previous studies of residential greenness, these studies were mostly based on North America and Europe. Our results of strong associations with increased buffer sizes indicate that the definitions of an effective area for residential greenness can vary by country, and future studies need to develop their definitions based on locally-specified behavioral characteristics.

There are several limitations to this study. First, the causal association between residential greenness and cortical thickness was not directly determined by our findings, as we used a cross-sectional study design. Future studies adopting a longitudinal design should investigate causality. Second, our exposure assessment was based on a single address at the time of the hospital visit, and we assumed constant exposure in the past without accounting for mobility. Future studies should include previous addresses to reduce exposure misclassification and to confirm our findings. Third, our study subjects had higher socioeconomic status than the general population in South Korea, given the high cost of medical examinations in a tertiary hospital. Future studies need to expand the population to represent the trends in a deprived population. Other socio-economic indicators such as income and occupation which were unavailable in our study also need to be applied in future studies.

5. Conclusions

Our study found that residential greenness in an urban population was positively associated with the cortical thickness of global, parietal and occipital regions. Although further studies with improved study designs are needed, this finding implies that the high exposure to residential greenness in an urban area may be beneficial for neurological disorders. Our finding of no association in the rural population suggests the need for a different interpretation of the impact of residential greenness depending on the type of area.

CRediT authorship contribution statement

KDM and JK drafted the manuscript. SWS and SYK supervised the project. KDM, JK, SWS and SYK developed study design and methodology. JK, YHP, HYS and CK contributed to data preparation. KDM, JK, YHP, HYS and CK conducted data analysis. All authors gave final approval to the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF; 2018R1A2B6004608) and was funded by the Research of Korea Center for Disease Control and Prevention (2018-ER6203-02). Additional support was provided by the National Cancer Center, Korea (NCC-1810220; NCC-2110570). In addition, this research was supported by a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare and Ministry of science and ICT, Republic of Korea (grant number: HU20C0111).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2021.146129.

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